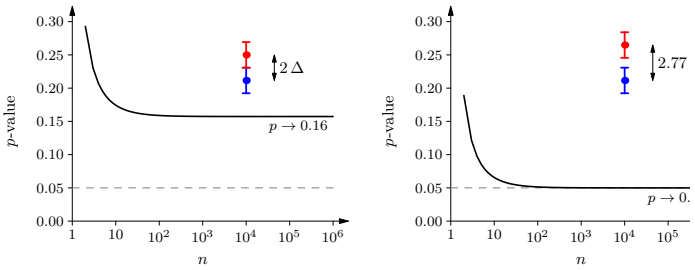
Mathematics of Machine Learning

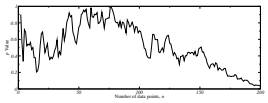
On Becoming a Scientist



interpreting data, standard errors, confidence tests

Outline

1. **Reporting Experiments**
2. Experimental Error
3. Confidence Estimates



Experiments and Reproducibility

- One of the fundamental requirements of science is that experiments can be reproduced
- This applies to machine learning and deep-learning as it does to particle physics, molecular biology, etc.
- We need to describe completely what procedures were carried out
- However, experiments have natural variability so what do we mean by repeatability?

Describing Research

- When describing research you need to find a story (narrative)—why is the work you are doing interesting
- You need to give a general motivation as well as a more detailed explanation of what you are specifically going to test
- You then need to give enough details that someone can take your report/paper and reproduce what you are doing
- These days it is also common practice to put the model you are using on-line—but you should still describe everything in your report

Describing Your Experiments

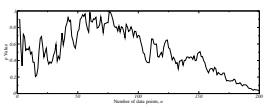
- You need to tell the reader
  - What your deep-learning architecture looks like
  - Your training set
  - How you partitioned your training set into train, validation and test sets
  - Any preprocessing you performed
  - The optimiser you used, mini-batch size, learning rate
  - The number of epochs you trained for
  - Any hyper-parameter tuning you carried out
  - How long it took in clock time on the platform you are using
- You can put some of this into an appendix if necessary

Results

- A paper builds a story (thesis) based on experimental evidences
- It is not enough to present results, it is also your job to interpret the results
- Your results provide evidence in support of your thesis
- But results all come with uncertainty. If you tested on a different test set, or used a different training set then the answer would be different
- **Results are meaningless unless the uncertainty is quantified!**

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Experimental Error

- Any empirical result is useless without some estimate of error
- If you tell me algorithm A has a performance of 95% and algorithm B has a performance of 97%, how do I know that this didn't happen by chance?
- Maybe algorithm A is actually better, but you got lucky (or unlucky)
- We need to get a handle on the estimated experimental error

Datasets

- We assume that our test set consists of typical examples that we might encounter and that they are independent (e.g. we are not seeing the same example over and over again)
- These are assumptions that are not always true
- We need to think about it and if they are not true you may need to collect new data or be very cautious of the claims that you are making
- However, if the person curating the training set did a good job then usually these are reasonable assumptions

Collecting Statistics

- Given some experimental measurements $\{X_i | i = 1,2,\ldots,n\}$ we can estimate the mean and variance

$$\hat{\mu}_n = \frac{1}{n}\sum_{i=1}^n X_i \qquad \hat{\sigma}_n^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \hat{\mu}_n)^2$$

- These are not the true mean, $\mu = \mathbb{E}[X_i]$, and true variance $\sigma^2 = \mathbb{E}\big[(X_i - \mu)^2\big]$, but they are unbiased estimates in that

$$\mathbb{E}[\hat{\mu}_n] = \mu \qquad \mathbb{E}\big[\hat{\sigma}_n^2\big] = \sigma^2$$

Variance in Mean Estimator

- Consider the mean squared error (MSE) of the estimator $\hat{\mu}_n$ (independent X_i)

$$\begin{aligned}
\mathbb{E}\big[(\hat{\mu}_n - \mu)^2\big] &= \mathbb{E}\left[\left(\left(\frac{1}{n}\sum_{i=1}^n X_i\right) - \mu\right)^2\right] = \mathbb{E}\left[\left(\frac{1}{n}\sum_{i=1}^n (X_i - \mu)\right)^2\right] \\
&= \frac{1}{n^2}\mathbb{E}\left[\left(\sum_{i=1}^n (X_i - \mu)\right)\left(\sum_{j=1}^n (X_j - \mu)\right)\right] \\
&= \frac{1}{n^2}\mathbb{E}\left[\sum_{i=1}^n (X_i - \mu)^2 + \sum_{i=1}^n \sum_{\substack{j=1 \\ j\neq i}}^n (X_i - \mu)(X_j - \mu)\right] \\
&= \frac{1}{n^2}\sum_{i=1}^n \left(\mathbb{E}\big[(X_i - \mu)^2\big] + \sum_{\substack{j=1 \\ j\neq i}}^n \mathbb{E}[X_i - \mu]\mathbb{E}[X_j - \mu]\right) \\
&= \frac{1}{n^2}\sum_{i=1}^n \sigma^2 = \frac{1}{n}\sigma^2
\end{aligned}$$

Estimated Error in the Mean

- The root mean squared error (RMSE) is equal to

$$\sqrt{\mathbb{E}[(\hat{\mu}_n - \mu)^2]} = \frac{\sigma}{\sqrt{n}}$$

- The **estimated error in the mean** or **standard error** is defined as $\Delta = \frac{\hat{\sigma}_n}{\sqrt{n}}$ where $\hat{\sigma}_n$ is the empirical estimate of the standard deviation
- Traditionally we would write an experimental result as $\hat{\mu}_n \pm \Delta$
- Also we only quote $\hat{\mu}_n$ up to the number of significant figures consistent with Δ
- Not 1.6582356 ± 0.122543 but 1.7 ± 0.1

Repeated Bernoulli Trials

- Suppose we are measuring a binary outcome, success or failure
- Assuming the trials are independent (so called Bernoulli trials) then the only unknown is the probability, p, of a success
- If we make n trials then probability of k successes is given by the binomial distribution

$$\mathbb{P}(K = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

- The mean and variance of K are

$$\mathbb{E}[K] = np \qquad \mathbb{E}\big[(K - np)^2\big] = np(1-p)$$

Mean Squared Error

- For the case of n Bernoulli trials we can estimate the mean success rate

$$\hat{p} = \frac{K}{n}$$

- The mean squared error (MSE) in this estimate is

$$\begin{aligned}
\mathbb{E}\big[(\hat{p} - p)^2\big] &= \mathbb{E}\left[\left(\frac{K}{n} - p\right)^2\right] = \frac{1}{n^2}\mathbb{E}\big[(K - np)^2\big] \\
&= \frac{1}{n^2} np(1-p) = \frac{p(1-p)}{n}
\end{aligned}$$

Standard Error for Bernoulli Trials

- The estimated error in the mean is given by the root mean squared error (RMSE) $\sqrt{\mathbb{E}[(\hat{p} - p)^2]} = \sqrt{\frac{p(1-p)}{n}}$
- We don't know p, but we can use an estimate, $\hat{p} = \frac{K}{n}$

$$\Delta \approx \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

- We can also use a worst-case bound (when $p = \frac{1}{2}$)

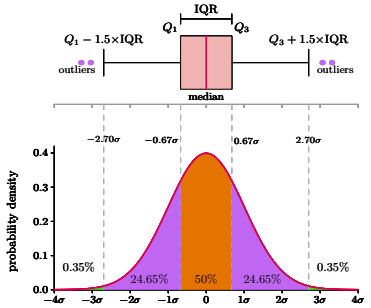
$$\Delta \approx \frac{1}{2\sqrt{n}}$$

This is pessimistic if p is small, although the previous estimate is optimistic if $\hat{p} = 0$

Errorbars

- There are different conventions on errorbars
- Traditionally errorbars show the estimated error in the mean (standard error, $\Delta = \hat{\sigma}_n/\sqrt{n}$)
- For roughly normal data the true mean is outside the errorbar about 32% of the time
- Some people use 95% confidence intervals as errorbars, saying 95% of the time the true mean lies within the errorbars—this corresponds to errorbars showing twice the standard error
- Some people show the standard deviation as errorbars, this tells you nothing about the confidence of the point you plot—in my opinion it is just wrong

Box Plots

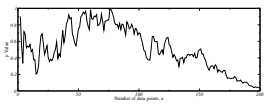


Box plots summarise the *spread* of the data.

They show no error bars on the median, so you cannot tell from them whether two results are statistically significantly different.

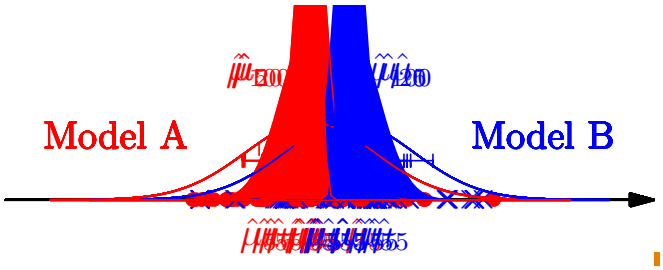
Outline

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Know What's Best

- How do we know if model B is better than model A?



The more data we have the less likely we will make a mistake

Quantifying Uncertainty

- We want a way of quantifying the chance that model B is better than model A
- To do this we can consider a **null hypothesis**

  We suppose that model A and B have the same performance
- We then ask, under the null hypothesis, what is the chance of obtaining a gap in performance at least as big as the one we observed
- This will depend on the number of observations and the empirical variance

T-tests

- The original t-test was invented by William Gosset who worked for the Guinness brewery in Dublin, but published under the pseudonym *Student*
- He assumed A and B are normally distributed with the same standard deviation; the null hypothesis is that their means are equal
- The more common test now is Welch's t-test, which drops the assumption that the standard deviations are equal
- The gap is equal to the t value

$$t = \frac{\hat{\mu}_A - \hat{\mu}_B}{\hat{\sigma}_{A,B}}$$

Welch's T-Test

- For Welch's t-test

$$t = \frac{\hat{\mu}_A - \hat{\mu}_B}{\hat{\sigma}_{A,B}} \qquad \hat{\sigma}_{A,B} = \sqrt{\frac{\hat{\sigma}_A^2}{n} + \frac{\hat{\sigma}_B^2}{m}}$$

- The p-value depends on t and the *number of degrees of freedom*

$$\nu = \frac{\hat{\sigma}_{A,B}^4}{(\hat{\sigma}_A^2/n)^2/(n-1) + (\hat{\sigma}_B^2/m)^2/(m-1)}$$

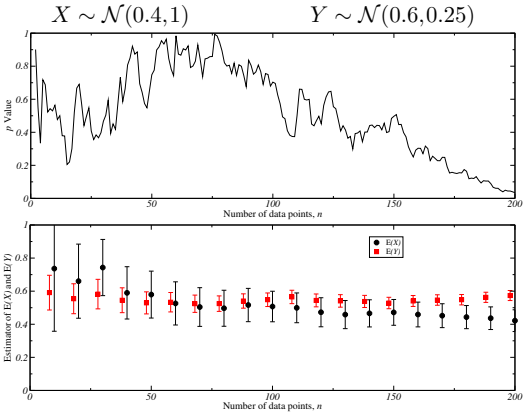
- The beauty of the t-test is that from t and ν we can compute p
- Small p is good. It shows the null hypothesis is unlikely

Statistical Significance

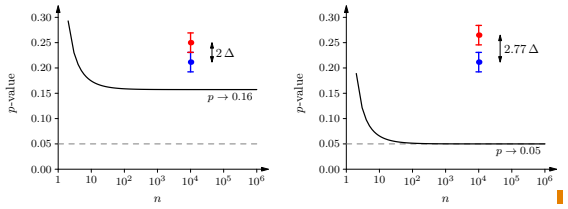
- It is common to say that A is *statistically significantly* better than B if, under the null hypothesis, the probability of a t at least as extreme as the one observed (the p-value) is less than 0.05
- Statistically significant difference in performance does not mean that the difference is anything to care about (that is a judgement call)
- We are assuming the data is normally distributed. We can check but no one does. Usually it's not a bad approximation if the data is not normal, but there are exceptions. . .

Example

$X \sim \mathcal{N}(0.4,1)$ $\quad$ $Y \sim \mathcal{N}(0.6,0.25)$



Error Bars and Significance



- Separate errorbars (1 standard error) are **not** enough: if they just touch, $p \approx 0.16$
- For $p < 0.05$ the gap must be about 2.8 standard errors

Paired T-Test

- Sometimes the fluctuations in performance will come both from the test example and from the difference in performance
- For example model A might always be better than model B, but the performance fluctuates depending on the example
- A normal t-test might not show any significant difference
- A paired t-test uses the per-example differences $X_i - Y_i$, whose standard error is

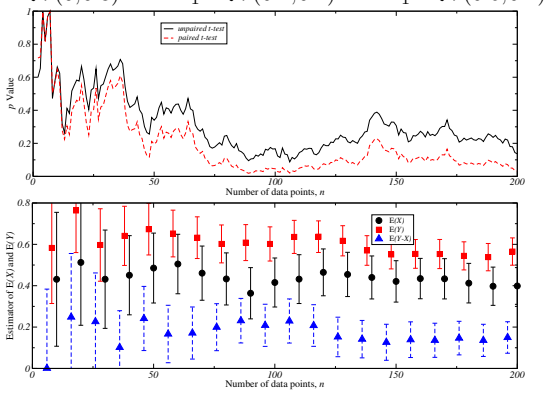
$$\hat{\sigma}^2_{X-Y} = \frac{\hat{\sigma}^2_X + \hat{\sigma}^2_Y - 2\hat{c}_{X,Y}}{n}$$

$$\hat{c}_{X,Y} = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \hat{\mu}_X)(Y_i - \hat{\mu}_Y)$$

Paired T-Test Example

$$X = Z + X_1 \qquad Y = Z + Y_1$$

$$Z \sim \mathcal{N}(0,0.8) \qquad X_1 \sim \mathcal{N}(0.4,0.2) \qquad Y_1 \sim \mathcal{N}(0.6,0.2)$$



Cheating Significance Tests

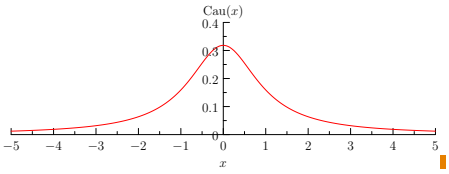
- You should only run your t-test once
- If you run multiple t-tests until you get one with $p < 0.05$ then you have cheated
- With a p value of 0.05 you expect to find the observed t value given the null hypothesis once every 20 trials—if you do the t-test twice you should expect this to happen every 10 trials (i.e. $p = 0.1$)
- If you run t-tests until you find a p value of 0.05 you are cheating
- If you want to compare many different models then use a method designed for it (e.g. ANOVA) or correct for the number of tests

Computing Significance Ab-Initio

- The t-test results follow from a beautiful, but difficult calculation
- However, you can simulate the null-hypothesis multiple times and measure empirically the number of times the t-value exceeds the observed t-values
- This isn't necessary under the assumptions that the data is normally distributed (done by Student and Welch)
- However, there are other times where you want to show that what you have found has not happened by chance. Often it is easy to simulate a null hypothesis

Good, Bad and the Ugly

- What I've told you is true of data that follows well behaved distributions
- Just occasionally you come across data that has long (thick) tails



- Here taking the average of many samples might not give you reliable estimates of the mean
- This isn't that common, but when nothing seems to make sense plot a histogram of the data and check for thick tails

Lessons

- You need to understand the errors in your data
- Computing the standard error is always necessary
- In some cases you can quantify your certainty that A is better than B using statistical confidence tests—t-tests (other tests are available)
- You need to apply common sense in interpreting data
- **The person you cheat when you don't do this is yourself**